

Developing a Fairness Auditing Tool for Income Prediction Models

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Introduction

With the increasing reliance on machine learning models for decision-making in critical areas such as finance and employment, it is imperative to ensure that these models operate fairly and equitably. Biases in ML models can lead to discriminatory practices, which are ethically and legally unacceptable. This project aims to develop a fairness auditing tool that detects and mitigates biases in income prediction models.

1- Objectives

- Identify and analyze the ethical implications of biases in income prediction models.
- Develop methodologies to detect and quantify biases related to race, gender, and other protected attributes.
- Explore and implement ethical strategies to mitigate identified biases in ML models.
- Evaluate the impact of bias mitigation strategies on ethical fairness and justice.
- Promote awareness and understanding of ethical issues in ML through comprehensive documentation and dissemination of findings.

2- Methodology

2-1 Data Collection and Preprocessing:

- Use the Adult Income Dataset.
- Preprocess the data to handle missing values, encode categorical variables, and normalize numerical features.

2-2 Model Development:

Train baseline income prediction models using algorithms such as logistic regression, decision trees, and neural networks.

2-3 Fairness Assessment:

- Implement fairness metrics such as demographic parity, equal opportunity, and disparate impact.
- Analyze the model outputs to detect biases against specific demographic groups.

2-4 Bias Mitigation:

- Apply techniques such as reweighting, adversarial debiasing, and fairness constraints during model training.
- Retrain models using these techniques and reassess their fairness.

2-5 Evaluation:

- Compare the performance of baseline and debiased models using both accuracy and fairness metrics.
- Conduct a thorough analysis to determine the trade-offs between model performance and fairness.

3- Questions to Address

- Identification of Bias: What are the main sources of bias in the Adult Income Dataset, and how do they impact the ethical fairness of income prediction models?
- Algorithmic Fairness: How do different ML algorithms compare in terms of fairness, and what are the ethical implications of these differences?
- Fairness Metrics: Which fairness metrics are most effective in detecting biases in income prediction models, and what ethical standards should guide their selection?
- Mitigation Techniques: What are the ethical impacts of various bias mitigation strategies on the fairness and accuracy of models, and how can we ensure they are implemented responsibly?
- Generalizability: How can we ensure that the fairness auditing solutions are ethically robust and generalizable to other datasets and applications, avoiding unintended consequences?